21/4/2026

Recall

DUIS:

$$I(\alpha) = \int_a^b f(x,\alpha)\,dx$$

one parameter

$$\frac{dI(\alpha)}{d\alpha} = \frac{d}{d\alpha}\left[\int_a^b f(x,\alpha)\,dx\right]$$

first operation → integration
2nd " → diff

By DUIS rule we can change the order of Differentiation & Integration is interchanged

DUIS tells me: we can differentiate first & then integrate

∴ By DUIS

$$\frac{dI}{d\alpha} = \int_a^b \left[\frac{\partial}{\partial\alpha} f(x,\alpha)\right] dx$$

HW

4. P.T $\displaystyle\int_0^\infty \frac{e^{-x}}{x}\left(e^{-ax}-e^{-bx}\right)dx = \log\frac{(1+b)}{(1+a)}$

Solⁿ: We have, the integrand

$$f(x,a,b) = \frac{e^{-x}}{x}\left(e^{-ax}-e^{-bx}\right)$$

Consider

$$\boxed{f_1(x,a) = \frac{e^{-x}}{x}e^{-ax}} \quad f_2(x,b) = \frac{e^{-x}}{x}e^{-bx}$$

Now, $\displaystyle I = I(a,b) = \int_0^\infty \left[\frac{e^{-x}}{x}\underbrace{e^{-ax}}_{1 \text{ at } 0} - \frac{e^{-x}}{x}\underbrace{e^{-bx}}_{1 \text{ at } 0}\right]dx$

Let $\cancel{I_1(a) = \int_0^\infty \frac{e^{-x}}{x}e^{-ax}\,dx}$

$$\Rightarrow I(a,b) = \int_0^\infty \left\{\left(\frac{e^{-x}}{x}-\frac{e^{-x}}{x}\right)+\frac{e^{-x}}{x}\left(e^{-ax}\right)-\frac{e^{-x}}{x}e^{-bx}\right\}dx$$

$$= \int_0^\infty \left\{\frac{e^{-x}}{x}\left(1-e^{-bx}\right)-\frac{e^{-x}}{x}\left(1-e^{-ax}\right)\right\}dx$$

4 (a). Evaluate $\int_0^\infty \frac{e^{-x}}{x}(1-e^{-bx})\,dx$

<u>Soln:</u> Let

$$I(b) = \int_0^\infty \frac{e^{-x}}{x}(1-e^{-bx})\,dx \quad \hookrightarrow (1)$$

Then by DUIS,

$$\frac{dI}{db} = \int_0^\infty \frac{\partial}{\partial b}\left[\frac{e^{-x}}{x}(1-e^{-bx})\right]dx$$

$$= \int_0^\infty \frac{e^{-x}}{x}\left[0-(-xe^{-bx})\right]dx$$

$$\Rightarrow \frac{dI}{db} = \int_0^\infty \frac{e^{-x}}{\cancel{x}}\left[\cancel{x}e^{-bx}\right]dx$$

ie $$\frac{dI}{db} = \int_0^\infty e^{-x(1+b)}\,dx$$

$$= \left.\frac{e^{-x(1+b)}}{-(1+b)}\right]_0^\infty$$

$$\Rightarrow \frac{dI}{db} = \frac{-1}{1+b}\left[0-1\right]$$

$$\Rightarrow \frac{dI}{db} = \frac{1}{1+b}$$

$$\Rightarrow \int dI = \int \frac{1}{1+b}\,db$$

$$\Rightarrow I = \log(1+b) + C \quad \hookrightarrow (2)$$

Put $b=0$ in (1):

$I(0) = 0$

$\Rightarrow$ from (2),

$I(0) = 0 = \log(1+0) + C$

$\Rightarrow 0 = \log 1 + C$

$\Rightarrow 0 = 0 + C$

$\Rightarrow \boxed{C = 0}$

$\therefore I = \log(1+b)$

ie $$\int_0^\infty \frac{e^{-x}}{x}(1-e^{-bx})\,dx = \log(1+b)$$

4. P.T $\int_0^\infty \frac{e^{-x}}{x}\left(e^{-ax}-e^{-bx}\right)dx = \log\left(\frac{1+b}{1+a}\right)$

Soln: Let

$$I = \int_0^\infty \frac{e^{-x}}{x}\left(e^{-ax}-e^{-bx}\right)dx$$

$$= \int_0^\infty \frac{e^{-x}}{x}\left(1-1+e^{-ax}-e^{-bx}\right)dx$$

$$= \int_0^\infty \frac{e^{-x}}{x}\left(\left(1-e^{-bx}\right)-\left(1-e^{-ax}\right)\right)dx$$

ie

$$I = \int_0^\infty \frac{e^{-x}}{x}\left(1-e^{-bx}\right)dx - \int_0^\infty \frac{e^{-x}}{x}\left(1-e^{-ax}\right)dx \quad (*)$$

$$= \log(1+b) - \log(1+a)$$

[Refer Q3 on 4(a)]

$$\Rightarrow I = \log\left(\frac{1+b}{1+a}\right)$$

5. Find $\int_0^\infty \frac{e^{-x}}{x}\left(a-\frac{1}{x}+\frac{1}{x}e^{-ax}\right)dx$

<u>Soln</u>: Let

$$I(a)=\int_0^\infty \frac{e^{-x}}{x}\left(a-\frac{1}{x}+\frac{1}{x}e^{-ax}\right)dx \quad (1)$$

$0-\frac{1}{x}+\frac{1}{x}\cdot 1$

By using DUIS,

$$\frac{dI}{da}=\int_0^\infty \frac{\partial}{\partial a}\left[\frac{e^{-x}}{x}\left(a-\frac{1}{x}+\frac{1}{x}e^{-ax}\right)\right]dx$$

$$=\int_0^\infty \frac{e^{-x}}{x}\left[1-0+\frac{1}{\cancel{x}}\left(-\cancel{x}e^{-ax}\right)\right]dx$$

$$\Rightarrow \frac{dI}{da}=\int_0^\infty \frac{e^{-x}}{x}\left(1-e^{-ax}\right)dx \quad (2)$$

Applying DUIS again, we get

$$\frac{d^2I}{da^2}=\int_0^\infty \frac{\partial}{\partial a}\left(\frac{e^{-x}}{x}\left(1-e^{-ax}\right)\right)dx$$

$$\Rightarrow \frac{d^2I}{da^2}=\int_0^\infty \frac{e^{-x}}{x}\left(0-(-x)e^{-ax}\right)dx$$

$$\Rightarrow \frac{d^2I}{da^2}=\int_0^\infty \frac{e^{-x}}{\cancel{x}}\left(\cancel{x}e^{-ax}\right)dx$$

$$\Rightarrow \frac{d^2I}{da^2}=\int_0^\infty e^{-x}\cdot e^{-ax}\,dx$$

ie $\dfrac{d^2I}{da^2} = \displaystyle\int_0^\infty e^{-x(1+a)}\,dx$

$$= \left.\frac{e^{-x(1+a)}}{-(1+a)}\right]_0^\infty$$

$$\Rightarrow \frac{d^2I}{da^2} = \frac{1}{1+a}$$

Integrating both sides w.r.t 'a' we get

$$\frac{dI}{da} = \int \frac{1}{1+a}\,da$$

$$\Rightarrow \frac{dI}{da} = \log(1+a) + c_1 \quad (3)$$

To find c_1:

In (2) put $a=0$. Then

$$\left.\frac{dI}{a}\right]_{a=0} = 0$$

Using this in (3), we get

$$\left.\frac{dI}{da}\right]_{a=0} = \log(1+0) + c_1$$

ie $0 = 0 + c_1$

$\Rightarrow c_1 = 0$

$$\therefore \frac{dI}{da} = \log(1+a)$$

Integrating both sides,

$$\int dI = \int \log(1+a)\,da$$

ie $I = (1+a)\log(1+a) - (1+a) + c_2 \quad (4)$

$\left[\int \log x\,dx = x\log x - x\right]$

Putting $a=0$ in (1) we get

$I(0) = 0$

Using this in (4), we get

$I(0) = 0 = (1+0)\log(1+0) - 1 + c_2$

$\Rightarrow c_2 = 1$

$\therefore I = (1+a)\log(1+a) - a$